PROOF OF A CONJECTURE OF SAIKIA AND TALUKDAR ON INTERNAL 2-ADIC CONGRUENCES FOR OVERPARTITION TUPLES

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ABSTRACT. Let $\overline{\mathrm{OPT}}_m(n)$ be the number of overpartition m-tuples of n into odd parts. For even m, Saikia and Talukdar reduced the internal congruences $\overline{\mathrm{OPT}}_m(2^i n) \equiv \overline{\mathrm{OPT}}_m(2^{i-1}n)$ to a conjectural pattern in the 2-adic valuations of a family of integer polynomials $R_i^{(m)}$ (their Conjecture 6.2), and they deduced congruences modulo 2^{3i+c_m} from it (their Corollary 6.3). We prove the conjecture for every even $m \geq 2$, in the full range $i > i_m$ in which it was stated ($i_m = 1$ if $4 \mid m$ and $i_m = 2$ otherwise). Writing ν for the 2-adic valuation, we also show that $\nu(R_i^{(m)}) = 3i-8$ if $m \equiv 2 \pmod 4$ and $\nu(R_i^{(m)}) = 3i+2\nu(m)-9$ if $4 \mid m$. Consequently $c_m = -4$ when $\nu(m) = 1$ and $c_m = 2\nu(m) - 5$ when $\nu(m) \geq 2$. This confirms the pattern in their data, that c_m depends only on $\nu(m)$, and it makes their Corollary 6.3 unconditional.

1. INTRODUCTION

For $m \geq 1$ let $\overline{\mathrm{OPT}}_m(n)$ denote the number of overpartition m-tuples of n whose parts are all odd. Its generating function is

$$\sum_{n\geq 0} \overline{\mathrm{OPT}}_m(n) q^n \;=\; A(q)^m, \qquad A(q) = \prod_{j\geq 1,\ j \text{ odd}} \frac{1+q^j}{1-q^j} = \frac{f_2^3}{f_1^2 f_4}, \qquad f_k = (q^k;q^k)_\infty.$$

Saikia and Talukdar [ST] proved, for every odd m and every $i \geq 3$, the congruence

$$\overline{\mathrm{OPT}}_m(2^i n) \equiv \overline{\mathrm{OPT}}_m(2^{i-1}n) \pmod{2^{i+1}},$$

and that the 2-adic valuation of the difference equals $i+1$ exactly at the odd squares. In [ST, §6] they treat even m through the following reduction, which we recall in their notation except that we write T_n for the Chebyshev polynomials of the first kind (they write D_n):

$$T_0 = 1, \qquad T_1 = x, \qquad T_{n+1} = 2xT_n - T_{n-1}.$$

The setting of [ST, §6]. Let $U\big(\sum c(n)q^n\big) = \sum c(2n)q^n$ and $\xi = A^4$. Then $\xi \in 1 + 8q\mathbb{Z}[[q]]$ by [ST, (2.16)] (see also Lemma 6.1), and in fact $\xi = 1 + 8q + O(q^2)$ because $A = 1 + 2q + O(q^2)$. Let $\mathcal{S}\colon \mathbb{Z}[x] \to \mathbb{Z}[x]$ be the $\mathbb{Z}$-linear map with $\mathcal{S}(x^j) = T_{2j}(x)$, and put

$$i_m = \begin{cases} 1, & 4 \mid m, \\ 2, & m \equiv 2 \pmod 4, \end{cases} \qquad P_{i_m - 1} = \begin{cases} x^{m/4}, & 4 \mid m, \\ T_{m/2}(x), & m \equiv 2 \pmod 4, \end{cases} \qquad P_{i+1} = \mathcal{S}(P_i).$$

Saikia and Talukdar show that $U^i(A^m) = P_i(\xi)$ for $i \geq i_m - 1$ [ST, §6, (6.3)–(6.4) and the display following (6.4)]; for completeness we include a short proof in Lemma 6.1. Consequently, with

$$R_i = R_i^{(m)} := \frac{P_i - P_{i-1}}{x-1} \in \mathbb{Z}[x] \qquad (i \geq i_m),$$

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the generating function of the differences is

$$\mathcal{D}_i^{(m)} := \sum_{n\geq 0} \big(\overline{\mathrm{OPT}}_m(2^i n) - \overline{\mathrm{OPT}}_m(2^{i-1}n)\big)q^n = (\xi - 1)\, R_i^{(m)}(\xi) \qquad (i \geq i_m), \tag{1}$$

which is [ST, (6.5)]. As in [ST, §4], for a nonzero integer c let $\nu(c)$ be its 2-adic valuation, and put $\nu(0) = \infty$. For $F \in \mathbb{Z}[x]$ let $\nu(F)$ be the minimum of ν over the coefficients of F in the monomial basis. Thus $\nu(F) \geq r$ if and only if $F \in 2^r\mathbb{Z}[x]$.

[ST, Lemma 6.1] introduces the operator

$$\widetilde{\Theta}(R) := \frac{\mathcal{S}\big((x-1)R\big)}{x-1} \qquad (R \in \mathbb{Z}[x]),$$

shows that $\widetilde{\Theta}(R) \in \mathbb{Z}[x]$ and $\widetilde{\Theta}(R)(1) = 4R(1) + 8R'(1)$, and shows that $R_{i+1} = \widetilde{\Theta}(R_i)$ for $i \geq i_m$.

Conjecture 1.1 ([ST, Conjecture 6.2])**.** *Let $m \geq 2$ be even, and put $v_i := \nu(R_i^{(m)})$. For every $i > i_m$:*

$$\nu\big((R_i^{(m)})'(1)\big) = v_i, \qquad \nu\big(R_i^{(m)}(1)\big) = v_i + 1, \qquad v_{i+1} = v_i + 3.$$

Assuming Conjecture 1.1, [ST, Corollary 6.3] states that for each even m there is an integer c_m with

$$\overline{\mathrm{OPT}}_m(2^i n) \equiv \overline{\mathrm{OPT}}_m(2^{i-1}n) \pmod{2^{3i+c_m}} \qquad (i > i_m,\ n \geq 0), \tag{2}$$

where $3i + c_m = v_i + 4$ and the modulus is attained at $n = 1$. Computations in [ST], for $2 \leq m \leq 14$ and $i \leq 9$ and carried out modulo 2^{45}, support the conjecture. They give $c_m = -4$ for $m \in \{2, 6, 10, 14\}$, $c_4 = c_{12} = -1$ and $c_8 = 1$. The authors note that these data suggest that c_m may depend only on $\nu(m)$, without claiming it.

Results.

Theorem 1.2. *Conjecture 1.1 holds for every even $m \geq 2$ and every $i > i_m$. Moreover, for $i > i_m$,*

$$v_i = \nu\big(R_i^{(m)}\big) = \begin{cases} 3i - 8, & m \equiv 2 \pmod 4, \\ 3i + 2\nu(m) - 9, & 4 \mid m. \end{cases}$$

Corollary 1.3. *Let $m \geq 2$ be even, and put*

$$c_m = \begin{cases} -4, & \nu(m) = 1, \\ 2\nu(m) - 5, & \nu(m) \geq 2. \end{cases}$$

Then the congruences (2) hold for all $i > i_m$ and $n \geq 0$, and $\nu\big(\overline{\mathrm{OPT}}_m(2^i) - \overline{\mathrm{OPT}}_m(2^{i-1})\big) = 3i + c_m$ exactly. In particular c_m is the constant of [ST, Corollary 6.3]*, and it depends only on $\nu(m)$.*

The corollary is unconditional. Its only inputs besides Theorem 1.2 are the identity (1) and $\xi \in 1 + 8q\mathbb{Z}[[q]]$, which do not depend on the conjecture: they are proved in [ST, §§2, 6], and again in Lemma 6.1 below.

For example, $(c_2, c_4, c_8, c_{16}, c_{32}, c_{64}) = (-4, -1, 1, 3, 5, 7)$. The first three values agree with the data of [ST].

Idea of the proof. We work in the variable $t = x - 1$. Section 3 gives closed forms for the images $\widetilde{\Theta}(t^k)$, obtained from the substitution $x = (y + y^{-1})/2$. They show that $\widetilde{\Theta}(t^k)$ is divisible by 2^{k+1} and has very few terms of degree < 2 in t. Consequently the pair $(R(1), R'(1))$ is transformed by an explicit linear recursion, modulo the right power of 2. In this form the conjectured pattern becomes an invariant of the recursion (Proposition 4.2), and it is started by explicit base cases (Section 5). A Lean 4 formalization is described in Appendix A.

2. Preliminaries

2.1. **Chebyshev polynomials.** Let y be an indeterminate. Since $(y+y^{-1})/2$ is transcendental over $\mathbb{Q}$, the assignment $x \mapsto (y+y^{-1})/2$ defines an injective ring homomorphism $\iota\colon \mathbb{Q}[x] \to \mathbb{Q}(y)$. We use two standard facts (see e.g. [MH]).

(T1) $\iota(T_n) = (y^n + y^{-n})/2$. This follows by induction from the recurrence, since $(y+y^{-1})(y^n + y^{-n}) - (y^{n-1} + y^{1-n}) = y^{n+1} + y^{-n-1}$.

(T2) $T_n(1) = 1$, $T_n'(1) = n^2$, and $T_n''(1) = n^2(n^2-1)/3$. By (T1), $T_n(\cos\theta) = \cos n\theta$, so $(1-x^2)T_n'' - xT_n' + n^2 T_n = 0$. Setting $x = 1$ gives $T_n'(1) = n^2 T_n(1) = n^2$. Differentiating once more and setting $x = 1$ gives $-3T_n''(1) + (n^2-1)T_n'(1) = 0$.

2.2. **The operators $\mathcal{S}$ and $\widetilde{\Theta}$.** Since $T_n(1) = 1$, we have $\mathcal{S}(F)(1) = F(1)$ for every $F \in \mathbb{Z}[x]$. In particular $\mathcal{S}((x-1)R)$ vanishes at $x = 1$, and as $x - 1$ is monic, $\widetilde{\Theta}(R) \in \mathbb{Z}[x]$. By linearity and (T1),

$$\iota\big(\mathcal{S}(F)\big) = \tfrac{1}{2}\big(F(y^2) + F(y^{-2})\big) \qquad (F \in \mathbb{Q}[x]), \tag{3}$$

where $F(y^{\pm 2})$ means F evaluated at $x = y^{\pm 2}$. Also, $R_{i+1} = \widetilde{\Theta}(R_i)$ for $i \geq i_m$, because $P_{i+1} - P_i = \mathcal{S}(P_i - P_{i-1}) = \mathcal{S}((x-1)R_i)$ (this is [ST, Lemma 6.1]).

2.3. **The variable $t = x-1$.** The substitution $F(x) \mapsto F(1+t)$ is a ring isomorphism $\mathbb{Z}[x] \to \mathbb{Z}[t]$ with inverse $G(t) \mapsto G(x-1)$. Both maps send $2^r\mathbb{Z}[\cdot]$ into $2^r\mathbb{Z}[\cdot]$. Hence, if we write

$$R(1+t) = \sum_{k \geq 0} r_k t^k,$$

then $\nu(R) = \min_k \nu(r_k)$. In other words, ν may be computed in the t-basis. Moreover $r_0 = R(1)$ and $r_1 = R'(1)$. From now on we identify $F \in \mathbb{Z}[x]$ with $F(1+t) \in \mathbb{Z}[t]$, and write $\widetilde{\Theta}(t^k)$ for $\widetilde{\Theta}((x-1)^k)$ expressed in t. Under ι,

$$\iota(t) = \iota(x-1) = \frac{(y-1)^2}{2y}, \qquad \iota(t+2) = \iota(x+1) = \frac{(y+1)^2}{2y}. \tag{4}$$

3. The action of $\widetilde{\Theta}$ on powers of t

Lemma 3.1. *For every $k \geq 0$, in $\mathbb{Q}(y)$,*

$$\iota\big(\widetilde{\Theta}(t^k)\big) = y\,(y-1)^{k-1}(y+1)^{k+1}\big(1 + (-1)^{k+1}y^{-2k-2}\big).$$

Proof. Take $F(x) = (x-1)^{k+1}$ in (3). Then $F(y^2) = (y^2-1)^{k+1}$ and $F(y^{-2}) = (y^{-2}-1)^{k+1} = (-1)^{k+1}y^{-2k-2}(y^2-1)^{k+1}$, so $\iota\big(\mathcal{S}((x-1)^{k+1})\big) = \frac{1}{2}(y^2-1)^{k+1}\big(1 + (-1)^{k+1}y^{-2k-2}\big)$. Dividing by $\iota(x-1) = (y-1)^2/(2y)$ gives the claim. □

Put $\omega := 2t(t+2) \in \mathbb{Z}[t]$, and for $l \geq 0$ let

$$W_l(u) := \frac{T_{l+1}(1+u) - T_l(1+u)}{u} \in \mathbb{Z}[u].$$

This is a polynomial with integer coefficients, because $T_{l+1}(1+u) - T_l(1+u) \in \mathbb{Z}[u]$ vanishes at $u = 0$ by (T2). For example $W_0 = 1$ and $W_1(u) = 3 + 2u$.

Lemma 3.2 (Closed forms)**.** *In $\mathbb{Z}[t]$ we have*

$$\widetilde{\Theta}(t^{2l-1}) = 4^l\, t^{l-1}(t+2)^l\, T_l(1+\omega) \qquad (l \geq 1),$$
$$\widetilde{\Theta}(t^{2l}) = 2 \cdot 4^l\, t^l(t+2)^{l+1}\, W_l(\omega) \qquad (l \geq 0).$$

Proof. By (4), $\iota(\omega) = (y^2-1)^2/(2y^2)$ and $\iota(1+\omega) = (y^2+y^{-2})/2$. Hence (T1) gives $\iota(T_l(1+\omega)) = (y^{2l}+y^{-2l})/2$. Since ι is injective, it suffices to compare images under ι with Lemma 3.1.

Odd exponents, $k = 2l-1$. The image of the right-hand side is

$$4^l \frac{(y-1)^{2l-2}}{(2y)^{l-1}} \frac{(y+1)^{2l}}{(2y)^l} \frac{y^{2l}+y^{-2l}}{2} = y^{1-2l}(y-1)^{2l-2}(y+1)^{2l}\big(y^{2l}+y^{-2l}\big)$$
$$= y(y-1)^{2l-2}(y+1)^{2l}\big(1+y^{-4l}\big).$$

This is Lemma 3.1 with $k = 2l-1$.

Even exponents, $k = 2l$. First,

$$\iota\big(T_{l+1}(1+\omega) - T_l(1+\omega)\big) = \tfrac{1}{2}\big(y^{2l+2}+y^{-2l-2}-y^{2l}-y^{-2l}\big)$$
$$= \tfrac{1}{2}(y^2-1)\big(y^{2l}-y^{-2l-2}\big),$$

so $\iota(W_l(\omega)) = y^2\big(y^{2l}-y^{-2l-2}\big)/(y^2-1)$. Hence the image of the right-hand side is

$$2\cdot 4^l \frac{(y-1)^{2l}(y+1)^{2l+2}}{(2y)^{2l+1}} \cdot \frac{y^2\big(y^{2l}-y^{-2l-2}\big)}{(y-1)(y+1)} = y(y-1)^{2l-1}(y+1)^{2l+1}\big(1-y^{-4l-2}\big).$$

This is Lemma 3.1 with $k = 2l$. □

Lemma 3.3. *(a) $\widetilde{\Theta}(t^k) \in 2^{k+1}\mathbb{Z}[t]$ for every $k \geq 0$.*

(b) We have $\widetilde{\Theta}(1) = 4+2t$ and $\widetilde{\Theta}(t) = 8+36t+32t^2+8t^3$. Moreover

$$\widetilde{\Theta}(t^2) \equiv 96t, \qquad \widetilde{\Theta}(t^3) \equiv 64t, \qquad \widetilde{\Theta}(t^k) \equiv 0 \quad (k \geq 4) \pmod{t^2}.$$

Proof. (a) In Lemma 3.2 the factors t, $t+2$, $T_l(1+\omega)$ and $W_l(\omega)$ have integer coefficients. The prefactor is $4^l = 2^{k+1}$ when $k = 2l-1$, and $2\cdot 4^l = 2^{k+1}$ when $k = 2l$.

(b) From Lemma 3.2:

- $\widetilde{\Theta}(1) = 2(t+2)$, using $W_0 = 1$.
- $\widetilde{\Theta}(t) = 4(t+2)(1+4t+2t^2) = 8+36t+32t^2+8t^3$.
- $\widetilde{\Theta}(t^2) = 8t(t+2)^2(3+2\omega) \equiv 8t\cdot 4\cdot 3 = 96t \pmod{t^2}$, using $W_1(u) = 3+2u$ and $\omega \equiv 0 \pmod{t}$.
- $\widetilde{\Theta}(t^3) = 16\,t(t+2)^2\,T_2(1+\omega) \equiv 16t\cdot 4\cdot T_2(1) = 64t \pmod{t^2}$.
- For $k = 2l-1 \geq 5$ the closed form contains the factor t^{l-1} with $l-1 \geq 2$. For $k = 2l \geq 4$ it contains t^l with $l \geq 2$. Either way $\widetilde{\Theta}(t^k) \equiv 0 \pmod{t^2}$. □

Corollary 3.4. *Let $R = \sum_k r_k t^k \in \mathbb{Z}[t]$ and write $\widetilde{\Theta}(R) = \sum_k r'_k t^k$. Then*

$$r'_0 = 4r_0 + 8r_1, \qquad r'_1 = 2r_0 + 36r_1 + 96r_2 + 64r_3.$$

Proof. Apply Lemma 3.3(b) to $\widetilde{\Theta}(R) = \sum_k r_k \widetilde{\Theta}(t^k)$. The first identity is the formula $\widetilde{\Theta}(R)(1) = 4R(1) + 8R'(1)$ of [ST, Lemma 6.1]. □

4. An invariant

Definition 4.1. Let $v \geq 0$. A polynomial $R = \sum_k r_k t^k \in \mathbb{Z}[t]$ (equivalently, an element of $\mathbb{Z}[x]$) is *good of level v* if

(G1) $\nu(R) = v$;
(G2) $\nu(r_1) = v$, that is, $\nu(R'(1)) = v$;
(G3) $\nu(r_0) = v+1$, that is, $\nu(R(1)) = v+1$;
(G4) $\alpha + \beta \equiv 2 \pmod 4$, where $\alpha := r_0/2^{v+1}$ and $\beta := r_1/2^v$.

By (G2) and (G3) the integers α and β are odd. Condition (G4) therefore says that $\alpha \equiv \beta \pmod 4$. If $R = R_i^{(m)}$ is good of level v, then $v = v_i$, and (G2), (G3) are the first two assertions of Conjecture 1.1 at the index i. The third assertion, $v_{i+1} = v_i + 3$, will follow from Proposition 4.2. Condition (G4) is the extra information needed to propagate them: by Corollary 3.4, the new constant term is $4r_0 + 8r_1 = 2^{v+3}(\alpha+\beta)$. Since α and β are odd, its valuation is at least $v+4$, and (G4) makes it exactly $v+4$. This is the cancellation between $4R(1)$ and $8R'(1)$ observed in [ST, §6].

Proposition 4.2 (Inductive step)**.** *If R is good of level v, then $\widetilde{\Theta}(R)$ is good of level $v+3$.*

Proof. Put $\rho_k := r_k/2^v$. These are integers by (G1), and $\rho_0 = 2\alpha$, $\rho_1 = \beta$. By Corollary 3.4,

$$r_0' = 4r_0 + 8r_1 = 2^{v+3}(\alpha+\beta),$$
$$r_1' = 2r_0 + 36r_1 + 96r_2 + 64r_3 = 2^{v+2}\gamma, \qquad \gamma := \alpha + 9\beta + 24\rho_2 + 16\rho_3.$$

Since $\alpha + \beta \equiv 2 \pmod 4$, we get $\nu(r_0') = v+4$. Since $\gamma \equiv \alpha + \beta \pmod 8$, we have $\nu(\gamma) = 1$ and hence $\nu(r_1') = v+3$.

Next we show that $\widetilde{\Theta}(R) \in 2^{v+3}\mathbb{Z}[t]$. Write $\widetilde{\Theta}(R) = r_0\widetilde{\Theta}(1) + r_1\widetilde{\Theta}(t) + \sum_{k\ge 2} r_k \widetilde{\Theta}(t^k)$. For $k \ge 2$, Lemma 3.3(a) gives $r_k\widetilde{\Theta}(t^k) \in 2^{v+k+1}\mathbb{Z}[t] \subseteq 2^{v+3}\mathbb{Z}[t]$. By Lemma 3.3(b),

$$r_0\widetilde{\Theta}(1) + r_1\widetilde{\Theta}(t) = (4r_0 + 8r_1) + (2r_0 + 36r_1)\,t + 32r_1\,t^2 + 8r_1\,t^3.$$

The coefficients here are $2^{v+3}(\alpha+\beta)$, $2^{v+2}(\alpha+9\beta)$, $2^{v+5}\beta$ and $2^{v+3}\beta$. They are all divisible by 2^{v+3}, because $\alpha + 9\beta$ is even. Hence $\nu(\widetilde{\Theta}(R)) \ge v+3$, and equality holds because $\nu(r_1') = v+3$. This proves (G1)–(G3) at level $v+3$.

Finally, $\alpha' = r_0'/2^{v+4} = (\alpha+\beta)/2$ and $\beta' = r_1'/2^{v+3} = \gamma/2$. Therefore

$$\alpha' + \beta' = \frac{2\alpha + 10\beta + 24\rho_2 + 16\rho_3}{2} = \alpha + 5\beta + 12\rho_2 + 8\rho_3 \equiv \alpha + \beta \equiv 2 \pmod 4,$$

which is (G4) at level $v+3$. □

Lemma 4.3 (Base case B1)**.** *If $R \in \mathbb{Z}[x]$ and $R(1)$ is odd, then $\widetilde{\Theta}(R)$ is good of level 1.*

Proof. By Corollary 3.4, $r_0' = 4(r_0 + 2r_1)$ and $r_1' = 2(r_0 + 18r_1 + 48r_2 + 32r_3)$. Since r_0 is odd, $\nu(r_0') = 2$ and $\nu(r_1') = 1$. By Lemma 3.3(a), $\widetilde{\Theta}(R) \in 2\mathbb{Z}[t]$, so $\nu(\widetilde{\Theta}(R)) = 1$. Finally,

$$\alpha' + \beta' = (r_0 + 2r_1) + (r_0 + 18r_1 + 48r_2 + 32r_3) \equiv 2r_0 \equiv 2 \pmod 4.$$ □

5. The base cases

Lemma 5.1 ($m \equiv 2 \pmod 4$)**.** *Let $m \equiv 2 \pmod 4$ and $n = m/2$, which is odd. Then $R_2^{(m)}(1) = n^2(4n^2+5)/3$, which is odd. Hence $R_3^{(m)}$ is good of level 1.*

Proof. Here $i_m = 2$ and $R_2 = (\mathcal{S}(T_n) - T_n)/(x-1)$. If $G \in \mathbb{Z}[x]$ and $G(1) = 0$, then $\big(G/(x-1)\big)(1) = G'(1)$. For $F = \sum_j f_j x^j$ we have

$$\mathcal{S}(F)'(1) = \sum_j f_j T_{2j}'(1) = 4\sum_j j^2 f_j = 4\big(F''(1) + F'(1)\big)$$

by (T2). With $F = T_n$ and (T2), this gives

$$R_2(1) = 4\Big(\frac{n^2(n^2-1)}{3} + n^2\Big) - n^2 = \frac{n^2(4n^2+5)}{3}.$$

The numerator is odd, so the integer $R_2(1)$ is odd. Lemma 4.3 applied to $R_3 = \widetilde{\Theta}(R_2)$ finishes the proof. □

Lemma 5.2 ($m \equiv 4 \pmod 8$)**.** *Let $m \equiv 4 \pmod 8$ and $n = m/4$, which is odd. Then $R_1^{(m)}(1) = n(4n-1)$, which is odd. Hence $R_2^{(m)}$ is good of level 1.*

Proof. Here $i_m = 1$ and $R_1 = (T_{2n} - x^n)/(x-1)$, so $R_1(1) = T'_{2n}(1) - n = 4n^2 - n$. Apply Lemma 4.3. □

Lemma 5.3 ($8 \mid m$; base case B2)**.** *Let $8 \mid m$. Put $n = m/4$, which is even, and $N = 2n = m/2$. Then $R_2^{(m)}$ is good of level $v := 2\nu(n) + 1 = 2\nu(m) - 3$.*

Proof. Put $Q := (T_N - 1)/(x-1)$ and $G := (x^n - 1)/(x-1)$, both in $\mathbb{Z}[x]$. Then $R_1 = (T_N - x^n)/(x-1) = Q - G$. Moreover

$$\widetilde{\Theta}(G) = \frac{\mathcal{S}(x^n - 1)}{x-1} = \frac{T_{2n} - 1}{x-1} = Q,$$

so $R_2 = \widetilde{\Theta}(R_1) = \widetilde{\Theta}(Q) - Q$.

Step 1: Q is good of level v. Write $Q(1+t) = \sum_k q_k t^k$. Since $Q(1+t) = (T_N(1+t) - 1)/t$, Taylor expansion and (T2) give

$$q_0 = T'_N(1) = N^2 = 4n^2, \qquad q_1 = \tfrac{1}{2}T''_N(1) = \frac{N^2(N^2-1)}{6} = \frac{2n^2(4n^2-1)}{3}.$$

Thus $\nu(q_0) = 2\nu(n) + 2 = v + 1$ and $\nu(q_1) = 2\nu(n) + 1 = v$.

To show $\nu(Q) \geq v$, put $F_M := T_M(1+t) - 1 \in \mathbb{Z}[t]$ for $M \geq 1$. The identity $T_{2M} = 2T_M^2 - 1$ follows from (T1), since $2\big((y^M + y^{-M})/2\big)^2 - 1 = (y^{2M} + y^{-2M})/2$. It gives

$$F_{2M} = 2F_M\,(F_M + 2).$$

Since $\nu(FG) \geq \nu(F) + \nu(G)$ for $F, G \in \mathbb{Z}[t]$, the bound $\nu(F_M) \geq c$ implies $\nu(F_{2M}) \geq 1 + c + \min(c, 1)$. We claim that $\nu(F_M) \geq 2\nu(M) - 1$ whenever $\nu(M) \geq 1$, and argue by induction on $\nu(M)$. If $M = 2M'$ with M' odd, then $\nu(F_{M'}) \geq 0$, so $\nu(F_M) \geq 1$. If $M = 2M'$ with $\nu(M') \geq 1$, then $c := 2\nu(M') - 1 \geq 1$ and $\nu(F_M) \geq c + 2 = 2\nu(M) - 1$. Since $F_N = t \cdot Q(1+t)$, we obtain $\nu(Q) = \nu(F_N) \geq 2\nu(N) - 1 = v$. Together with $\nu(q_1) = v$, this gives $\nu(Q) = v$.

It remains to check (G4) for Q. Write $n = 2^{\nu(n)}n'$ with n' odd. Then $\alpha_Q = q_0/2^{v+1} = n'^2 \equiv 1 \pmod 8$ and $\beta_Q = q_1/2^v = n'^2 \cdot (4n^2-1)/3$. Since n is even, $3 \cdot \frac{4n^2-1}{3} \equiv -1 \pmod{16}$, so $(4n^2-1)/3 \equiv 5 \pmod{16}$. Hence $\beta_Q \equiv 5 \pmod 8$, and $\alpha_Q + \beta_Q \equiv 6 \equiv 2 \pmod 4$. So Q is good of level v.

Step 2. By Proposition 4.2, $\widetilde{\Theta}(Q)$ is good of level $v + 3$. In particular $\widetilde{\Theta}(Q) \in 2^{v+3}\mathbb{Z}[t]$, and therefore

$$R_2 \equiv -Q \pmod{2^{v+3}\mathbb{Z}[t]}.$$

Since $v+1 < v+3$, it follows that $\nu(R_2) = \nu(Q) = v$, $\nu(r_1) = \nu(q_1) = v$ and $\nu(r_0) = \nu(q_0) = v+1$, where $R_2(1+t) = \sum r_k t^k$. Furthermore $\alpha_{R_2} \equiv -\alpha_Q \pmod 4$ and $\beta_{R_2} \equiv -\beta_Q \pmod 8$, so $\alpha_{R_2} + \beta_{R_2} \equiv -2 \equiv 2 \pmod 4$. Hence R_2 is good of level v. □

6. Proofs of Theorem 1.2 and Corollary 1.3

The theorem concerns only the polynomials $R_i^{(m)}$. To pass to the congruences we need the q-series identity (1), which is [ST, (6.5)]. To keep the note self-contained we include a short proof of it. Let $\varphi(q) = \sum_{k \in \mathbb{Z}} q^{k^2}$ and $(a;q)_\infty = \prod_{j \geq 0}(1 - aq^j)$.

Lemma 6.1 (cf. [ST, (2.16), (6.3)])**.** *(a) $A^2 = \varphi(q)/\varphi(-q)$, and $\xi = A^4 \in 1 + 8q\mathbb{Z}[[q]]$.*

(b) $U(A^{2k}) = T_k(\xi)$ for every $k \geq 0$. Consequently $U\big(F(\xi)\big) = \mathcal{S}(F)(\xi)$ for every $F \in \mathbb{Z}[x]$.

(c) For every even $m \geq 2$ and $i \geq i_m - 1$ we have $U^i(A^m) = P_i(\xi)$. Hence (1) holds for $i \geq i_m$.

Proof. (a) By the Jacobi triple product identity [And, Theorem 2.8], $\varphi(q) = (q^2;q^2)_\infty(-q;q^2)_\infty^2$, and replacing q by $-q$ gives $\varphi(-q) = (q^2;q^2)_\infty(q;q^2)_\infty^2$. Since $A = (-q;q^2)_\infty/(q;q^2)_\infty$, this gives $A^2 = \varphi(q)/\varphi(-q)$. Next, $\varphi(q) - \varphi(-q) = 4\sum_{k\geq 1,\ k \text{ odd}} q^{k^2} \in 4q\mathbb{Z}[[q]]$, and $\varphi(-q) \in 1 + q\mathbb{Z}[[q]]$ is a unit of $\mathbb{Z}[[q]]$. Hence $A^2 - 1 \in 4q\mathbb{Z}[[q]]$, so $A^2 + 1 \in 2\mathbb{Z}[[q]]$ and $\xi - 1 = (A^2-1)(A^2+1) \in 8q\mathbb{Z}[[q]]$.

(b) We use two classical identities:

$$\varphi(q)\varphi(-q) = \varphi(-q^2)^2, \qquad \varphi(q)^2 + \varphi(-q)^2 = 2\varphi(q^2)^2. \tag{5}$$

For the first, note that $(-q;q^2)_\infty(q;q^2)_\infty = (q^2;q^4)_\infty$ and $(q^2;q^2)_\infty = (q^2;q^4)_\infty(q^4;q^4)_\infty$. By the product formulas in (a), both sides then equal $(q^4;q^4)_\infty^2(q^2;q^4)_\infty^4$. For the second, write $\varphi(q)^2 = \sum_n r(n)q^n$, where $r(n)$ is the number of $(a,b) \in \mathbb{Z}^2$ with $a^2+b^2 = n$. Then $\varphi(q)^2 + \varphi(-q)^2 = 2\sum_n r(2n)q^{2n}$. If $a^2+b^2 = 2n$ then $a \equiv b \pmod 2$, and $(a,b) \mapsto \left(\frac{a+b}{2}, \frac{a-b}{2}\right)$ is a bijection onto the solutions of $c^2+d^2 = n$, with inverse $(c,d) \mapsto (c+d, c-d)$. So $r(2n) = r(n)$.

Put $X = A^2$, viewed in the field $\mathbb{Q}((q))$. For $F \in \mathbb{Z}[[q]]$ we have $U(F)(q^2) = \frac{1}{2}\big(F(q) + F(-q)\big)$. By (a), $X(-q) = X(q)^{-1}$. By (T1), the identity $T_k\big((z+z^{-1})/2\big) = (z^k + z^{-k})/2$ holds in $\mathbb{Q}[z,z^{-1}]$, and we may specialize it to $z = X$. Therefore

$$U(X^k)(q^2) = \frac{X^k + X^{-k}}{2} = T_k\Big(\frac{X + X^{-1}}{2}\Big),$$

and by (5)

$$\frac{X + X^{-1}}{2} = \frac{\varphi(q)^2 + \varphi(-q)^2}{2\varphi(q)\varphi(-q)} = \frac{\varphi(q^2)^2}{\varphi(-q^2)^2} = X(q^2)^2 = \xi(q^2).$$

Hence $U(A^{2k})(q^2) = T_k(\xi(q^2))$, that is, $U(A^{2k}) = T_k(\xi)$. Taking $k = 2j$ gives $U(\xi^j) = T_{2j}(\xi)$, and the second assertion follows by linearity of U.

(c) If $4 \mid m$, then $U^0(A^m) = \xi^{m/4} = P_0(\xi)$. If $m \equiv 2 \pmod 4$, then (b) with $k = m/2$ gives $U(A^m) = T_{m/2}(\xi) = P_1(\xi)$. In both cases the start is $i = i_m - 1$, and (b) gives $U(P_i(\xi)) = \mathcal{S}(P_i)(\xi) = P_{i+1}(\xi)$. Finally, for $i \geq i_m$,

$$\mathcal{D}_i^{(m)} = U^i(A^m) - U^{i-1}(A^m) = P_i(\xi) - P_{i-1}(\xi) = (\xi - 1)R_i^{(m)}(\xi). \qquad \square$$

Proof of Theorem 1.2. Let $i_0 := i_m + 1$. By Lemmas 5.1, 5.2 and 5.3, R_{i_0} is good of some level λ_m:

- if $m \equiv 2 \pmod 4$, then $i_0 = 3$ and $\lambda_m = 1 = 3\cdot 3 - 8$;
- if $m \equiv 4 \pmod 8$, then $i_0 = 2$ and $\lambda_m = 1 = 3\cdot 2 + 2\cdot 2 - 9$;
- if $8 \mid m$, then $i_0 = 2$ and $\lambda_m = 2\nu(m) - 3 = 3\cdot 2 + 2\nu(m) - 9$.

Since $R_{i+1} = \widetilde{\Theta}(R_i)$ for $i \geq i_m$, Proposition 4.2 and induction show that R_i is good of level $\lambda_m + 3(i - i_0)$ for every $i \geq i_0$. This level equals the value of v_i claimed in the theorem. For each $i > i_m$, condition (G1) gives $v_i = \nu(R_i)$, (G2) gives $\nu(R_i'(1)) = v_i$, and (G3) gives $\nu(R_i(1)) = v_i + 1$. The equality $v_{i+1} = v_i + 3$ is the shift of the level in Proposition 4.2. $\square$

Proof of Corollary 1.3. We recall the argument of [ST, proof of Cor. 6.3], which uses only (1), $\xi \in 1 + 8q\mathbb{Z}[[q]]$ (both proved in Lemma 6.1) and the three assertions of Conjecture 1.1. Fix $i > i_m$ and write $R_i = 2^{v_i}R_i^*$ with $R_i^* \in \mathbb{Z}[x]$. By Theorem 1.2, $R_i^*(1) = R_i(1)/2^{v_i} \equiv 2 \pmod 4$. Since $R_i^*(x) - R_i^*(1) \in (x-1)\mathbb{Z}[x]$, we get $R_i^*(\xi) - R_i^*(1) \in (\xi - 1)\mathbb{Z}[[q]] \subseteq 8\mathbb{Z}[[q]]$. So $R_i^*(\xi) \in 2\mathbb{Z}[[q]]$, and by (1) $\mathcal{D}_i^{(m)} = 2^{v_i}(\xi - 1)R_i^*(\xi) \in 2^{v_i+4}\mathbb{Z}[[q]]$. Moreover, since $\xi - 1 = 8q + O(q^2)$, the coefficient of q in $\mathcal{D}_i^{(m)}$ equals $8R_i(1)$, which has valuation exactly $v_i + 4$. Thus (2) holds with $3i + c_m = v_i + 4$, and the modulus is attained at $n = 1$. Inserting the values of v_i from Theorem 1.2 gives $c_m = (3i-8)+4-3i = -4$ when $m \equiv 2 \pmod 4$, and $c_m = (3i + 2\nu(m) - 9) + 4 - 3i = 2\nu(m) - 5$ when $4 \mid m$. $\square$

7. Remarks

Remark 7.1 (The hypothesis $i > i_m$). The hypothesis $i > i_m$ in Conjecture 1.1 cannot be dropped in general. Let $i = i_m$. Then $v_{i_m} = \nu(R_{i_m}) = 0$ in all cases. For $m \not\equiv 0 \pmod 8$ this holds because $R_{i_m}(1)$ is odd by Lemmas 5.1 and 5.2. For $8 \mid m$ it holds because $R_1 = Q - G$, where $\nu(Q) \geq 3$ and $G = 1 + x + \cdots + x^{n-1}$.

- If $m \not\equiv 0 \pmod 8$, then $R_{i_m}(1)$ is odd, so $\nu(R_{i_m}(1)) = 0 \neq v_{i_m} + 1$. For example, $R_2^{(2)} = 2x+1$, and $R_1^{(12)} = (T_6 - x^3)/(x-1)$ satisfies $R_1^{(12)}(1) = 33$. Moreover $v_{i_m+1} = 1 \neq v_{i_m} + 3$.
- If $16 \mid m$, then $R_1'(1) = q_1 - n(n-1)/2$, and $\nu(q_1) = 2\nu(n) + 1 > \nu(n) - 1 = \nu\big(n(n-1)/2\big)$. Hence $\nu(R_1'(1)) = \nu(m) - 3 \geq 1 \neq v_1$. For example, $m = 16$ gives $(\nu(R_1), \nu(R_1'(1)), \nu(R_1(1))) = (0, 1, 2)$.
- If $m \equiv 8 \pmod{16}$, the same computation gives $\nu(R_1'(1)) = 0$ and $\nu(R_1(1)) = \nu(4n^2 - n) = 1$, and $v_2 = 3$ by Lemma 5.3. So all three assertions happen to hold at $i = i_m$ as well.

A computation for all even $m \leq 400$ confirms this trichotomy. The congruence (2) itself can also fail at $i = i_m$. Since $A = 1 + 2q + 2q^2 + O(q^3)$, we have $\overline{\mathrm{OPT}}_m(1) = 2m$ and $\overline{\mathrm{OPT}}_m(2) = 2m^2$. Hence $\overline{\mathrm{OPT}}_m(2) - \overline{\mathrm{OPT}}_m(1) = 2m(m-1)$ has valuation $\nu(m) + 1$. For $4 \mid m$ this is smaller than $3i_m + c_m = 2\nu(m) - 2$ exactly when $16 \mid m$. For instance $\nu(\overline{\mathrm{OPT}}_{16}(2) - \overline{\mathrm{OPT}}_{16}(1)) = \nu(480) = 5 < 6$.

Remark 7.2. The proof gives slightly more than the conjecture asks for. For $i > i_m$, the normalized values $\alpha_i = R_i(1)/2^{v_i+1}$ and $\beta_i = R_i'(1)/2^{v_i}$ are odd and satisfy $\alpha_i \equiv \beta_i \pmod 4$. Moreover $\alpha_{i+1} = (\alpha_i + \beta_i)/2$. The value of c_m is read off from the base case: if R_{i_0} is good of level λ_m, then $v_i = \lambda_m + 3(i - i_0)$, so $c_m = \lambda_m - 3i_0 + 4$. For $m \equiv 2 \pmod 4$ the base case is R_3, of level 1, which gives $c_m = -4$. For $4 \mid m$ it is R_2, of level $2\nu(m) - 3$, which gives $c_m = 2\nu(m) - 5$. (When $m \equiv 8 \pmod{16}$, one can check that R_1 is already good of level 0, in line with Remark 7.1. This gives the same value of c_m.)

8. Computational verification

The proof above is complete and does not rely on computation. As a safeguard, all statements were also checked by computer. The checks used several programs, written separately and using exact integer arithmetic, or arithmetic modulo 2^K with K far above every valuation involved. The software was PARI/GP 2.15.4 [PARI] and Python 3 with python-flint 0.9.0 [FLINT] and NumPy.

1. *The conjecture and the formula for v_i, from the definitions of* [ST, §6]*:*
   - all even $m \leq 400$ with $i \in \{i_m + 1, i_m + 2\}$ (PARI/GP);
   - all even $m \leq 200$ with $i \leq i_m + 5$ (exact modulo 2^{160}, with $\mathcal{S}$ evaluated by Clenshaw summation);
   - all even $m \leq 200$ with $i \leq i_m + 6$, and all even $m \leq 100$ with $i \leq i_m + 8$ (modulo 2^{64}, an independent implementation);
   - all even $m \leq 200$ with $i \leq i_m + 3$ (exact big integers, python-flint; this check also confirms $R_{i+1} = \widetilde{\Theta}(R_i)$ exactly);
   - $m \in \{2^k : k \leq 11\} \cup \{3 \cdot 2^k : k \leq 9\} \cup \{5 \cdot 2^k : k \leq 8\}$ with $i \leq i_m + 3$ (modulo 2^{64}), which covers $\nu(m)$ up to 11.

   No discrepancy was found.
2. *The lemmas:*
   - the closed forms of Lemma 3.2 and Lemma 3.3(b), for $k \leq 60$;
   - Lemma 3.3(a), for $k \leq 80$; in fact the exact content of $\widetilde{\Theta}(t^k)$ is 2^{k+1};
   - the parity statements of Lemmas 5.1 and 5.2, for $m \leq 2400$;
   - Lemma 5.3 (including the residues $\alpha_Q \equiv 1$, $\beta_Q \equiv 5 \pmod 8$), for $8 \mid m \leq 1200$.

3. *The congruences themselves, directly from the generating function.* We used the identity $A^2 = \varphi(q)/\varphi(-q)$, where $\varphi(q) = \sum_{k\in\mathbb{Z}} q^{k^2}$. We verified $\nu\big(\overline{\mathrm{OPT}}_m(2^i n) - \overline{\mathrm{OPT}}_m(2^{i-1}n)\big) \geq 3i + c_m$, with equality at $n = 1$, in the following ranges:
   - all even $m \leq 128$, $i_m < i \leq 14$ and $2^i n \leq 16384$ (800 pairs (m, i));
   - separately, from the eta-quotient $f_2^3/(f_1^2 f_4)$, for even $m \leq 16$, $i \leq 7$ and $2^i n \leq 6000$.

All scripts, together with the Lean source file of Appendix A, are included as ancillary files with the arXiv version of this note (directory `anc/`).

AI usage statement

The research presented here is the result of human–AI interaction involving Claude (Anthropic). The author selected the problem, directed the work, and decided what to check and how to present it. The proof of Theorem 1.2, including the change of variable $t = x - 1$, the closed forms of Section 3, the invariant of Section 4 and the base cases, was found by the AI system in the course of this interaction. The same holds for the proof of Lemma 6.1, the verification programs of Section 8 and the Lean formalization of Appendix A. The manuscript was drafted with the AI system and revised under the author's direction.

The results were checked in three independent ways: by separate AI review passes that re-derived each step, by the computations of Section 8 (programs written independently of one another, see the ancillary files), and by the Lean 4 formalization of Theorem 1.2, which uses no unproved assumptions. The note has not yet been reviewed by a human specialist in the area. Conjecture 1.1 was posed in [ST] as open, supported by computations for $m \leq 14$, and to the author's knowledge it was unsolved before this note. The author takes full responsibility for the content and welcomes detailed questions about the role of AI in this work.

Appendix A. A Lean 4 formalization

Theorem 1.2 has been formalized and machine-checked in Lean 4 (toolchain `v4.35.0-rc4`) with Mathlib (the matching tag `v4.35.0-rc4`), in a single file `SaikiaTalukdar.lean`, which is included as the ancillary file `anc/lean/SaikiaTalukdar.lean`. The definitions follow [ST, §6] literally:

- the Chebyshev polynomials $D_n \in \mathbb{Z}[x]$, by the three-term recurrence;
- the $\mathbb{Z}$-linear operator $\mathcal{S}$ with $\mathcal{S}(x^j) = D_{2j}$;
- i_m and P_{i_m-1} as in the introduction, with $P_i = \mathcal{S}^{\,i-i_m+1}(P_{i_m-1})$;
- $R_i = (P_i - P_{i-1})/(x-1)$, computed as division by the monic polynomial $x - 1$;
- $\nu(F)$, the minimum of the 2-adic valuations of the nonzero coefficients of F.

The file proves the following.

- `conjecture_6_2`: for every even $m \geq 2$ and every $i > i_m$, $R_i \neq 0$, $R_i(1) \neq 0$ and $R_i'(1) \neq 0$; moreover $\nu(R_i'(1)) = \nu(R_i)$, $\nu(R_i(1)) = \nu(R_i) + 1$ and $\nu(R_{i+1}) = \nu(R_i) + 3$.
- `nu_R_of_mod4_eq2` and `nu_R_of_four_dvd`: the two formulas for v_i in Theorem 1.2.

The formal proof follows Sections 3–6: the t-coordinate, the bounds of Lemma 3.3, the invariant of Section 4, and the three base cases, including the content argument based on $F_{2M} = 2F_M(F_M+2)$. The only difference is that the closed forms of Lemma 3.2 are obtained by a forward-difference argument rather than the substitution $x = (y + y^{-1})/2$. For each of the three theorems, the command `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. In particular, no `sorry` is used. Corollary 1.3 depends on the q-series identities of [ST], and it is not formalized.

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